DO NOW — RECALL

Paper 2 examiner-style feedback

Name: _____ Class: _____ Date: _____

1. Why must recursive calls progress?

2. What does a midpoint calculate?

3. What happens when low>high?

4. C# [2,4,6] with a[1]=a[0]+a[2]. Final array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

To reach the base case.

2. Model answer

The middle index of remaining interval.

3. Model answer

No candidates remain; not found.

4. Model answer

[2,8,6].

By the end, you can…

• Identify whether the error lies in problem reading, algorithm state or pseudocode.

• Explain every bound update, recursive return or data-structure operation.

• Retest the method with a different input to demonstrate transfer.

Vocabulary — know these terms

mark scheme — Criteria for awarding marks.

termination — Algorithm eventually stops.

auxiliary space — Additional storage beyond the input.

NEW LEARNING

Read, annotate and explain

Understand

Identify whether the error lies in problem reading, algorithm state or pseudocode.

Apply

Explain every bound update, recursive return or data-structure operation.

Evaluate

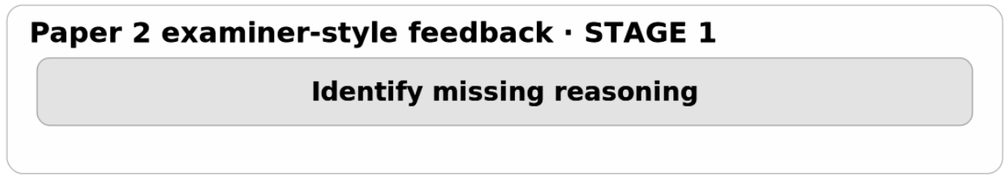
Retest the method with a different input to demonstrate transfer.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

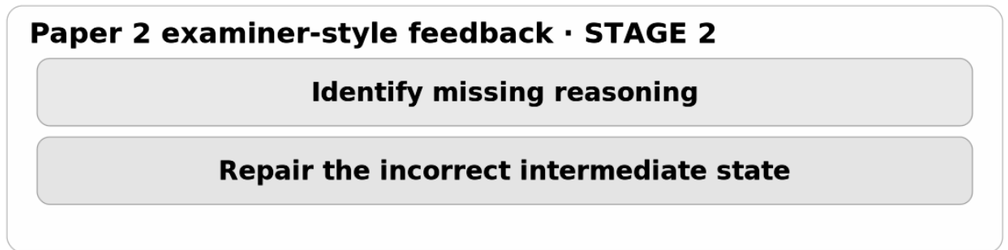
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Paper 2 examiner-style feedback · STAGE 1

New Learning



Paper 2 examiner-style feedback · STAGE 2

New Learning



Paper 2 examiner-style feedback · STAGE 3

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Repair a binary search that can repeat the midpoint forever.

The boundaries exclude a failed midpoint, so the remaining interval shrinks.



WEDO · METHOD

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

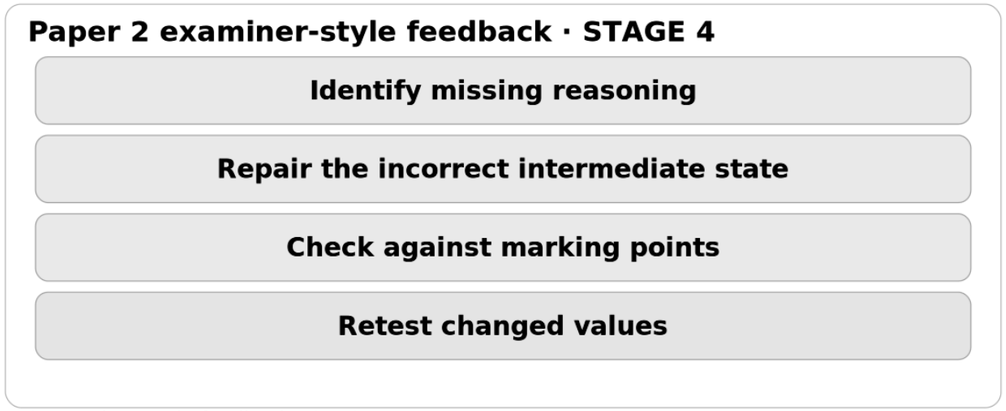
Guided 1 — CHECK

Explain the result
The boundaries exclude a failed midpoint, so the remaining interval shrinks.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Paper 2 examiner-style feedback · STAGE 4

I DO — WORKED EXAMPLE

Follow the worked method

Repair a SumTo recursion missing a base case.

Explain the result

The n==0 base case stops recursion and returns zero.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Getting a correct output once guarantees an algorithm is bug-free.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Correct: 'merge sort has O(1) auxiliary space because the temp array is short-lived'. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Repair a SumTo recursion missing a base case.

The n==0 base case stops recursion and returns zero.



IDO · METHOD

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Repair a binary search that can repeat the midpoint forever.
Source code / input:
if(a[mid]<target)low=mid+1;
else if(a[mid]>target)high=mid-1;

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

An algorithm's recursion has no base case. Explain fault and give two repairs. [6]

Question 2

Trace binary search for 9 in [1,3,5,7,9], recording every low, high and midpoint. [6]